

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Paper No. BPT-503: Quantum Mechanics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer any **four** of the following:

[4 × 5 = 20]

(a) Find the de Broglie wavelength associated with:

(i) an electron accelerated through a potential difference of 10 MV.

(ii) a proton having a kinetic energy of 10 MeV.

(Take rest mass energy of electron = 0.51 MeV and rest mass energy of proton = 938 MeV).

(b) Using the uncertainty relation, explain why electrons cannot exist inside the nucleus.

(c) What are stationary states? Explain with reference to the time-dependent and time-independent Schrödinger equations.

(d) Prove the statement: “commuting observables have simultaneous eigenstates”.

(e) Evaluate the following commutation relations:

(i) $[L_x, y]$ (ii) $[L_z, p_x]$ (f) Find the most probable value of the radial distance r of the electron in the ground state of the Hydrogen atom.**Question 2.**

(a) Describe the difference between phase velocity and group velocity for a wave packet. [4]

(b) A particle is constrained to move in a potential of the form:

$$V(x) = \begin{cases} 0, & 0 < x < a \\ \infty, & \text{otherwise} \end{cases}$$

Obtain the values of energy E and corresponding normalized eigenfunctions. Show that the eigenfunctions are mutually orthogonal. Is there any acceptable physical solution with $E = 0$ or $E < 0$? Justify your answer. [10]

Question 3.

- (a) Describe the Davisson-Germer experiment to demonstrate the wave nature of the electron. [4]
- (b) If a particle having energy $E > 0$ is incident from the left upon a one-dimensional square potential well defined as:

$$V(x) = \begin{cases} -V_0, & -a < x < a \\ 0, & \text{otherwise} \end{cases}$$

find the transmission coefficient T and plot its variation qualitatively as a function of the ratio E/V_0 . [10]

OR

- (a) Using the Schwarz inequality, give the mathematical proof of the uncertainty relation for canonically conjugate pairs. [6]
- (b) Find the wave function $\psi(x)$, which is the Fourier transform of $\phi(k)$, for a wave packet whose amplitude function $\phi(k)$ varies as:

$$\phi(k) = \begin{cases} g_0, & -K_0 < k < K_0 \\ 0, & \text{otherwise} \end{cases}$$

Calculate the probability of finding the particle in the region $-\frac{\pi}{K_0} < x < \frac{\pi}{K_0}$. [8]

Question 4.

- (a) What is the difficulty in normalizing a free particle wave function? How is the concept of a wave packet used to resolve this difficulty? [5]
- (b) Find the uncertainty in position and momentum for the Gaussian wave packet $\psi(x) = Ae^{-ax^2}$. How does it obey the Heisenberg uncertainty relation? [9]

OR

- (a) The solution of the Schrödinger equation for a free particle in one dimension is of the form:

$$\psi(x) = A \sin(kx) + B \cos(kx)$$

where A and B are complex constants.

- (i) Calculate the probability current density for this wave function. [4]
- (ii) Find the relation between the coefficients A and B , if $\psi(x)$ is an eigenfunction of the momentum operator. [4]
- (b) Show that the Harmonic Oscillator Hamiltonian can be written as:

$$H = \hbar\omega \left(a^\dagger a + \frac{1}{2} \right)$$

in terms of the lowering and raising operators:

$$a = \sqrt{\frac{m\omega}{2\hbar}} \left(x + \frac{ip}{m\omega} \right), \quad a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(x - \frac{ip}{m\omega} \right)$$

Work out the action of the operators a and $a^\dagger$ on the energy eigenstates of the Hamiltonian. [6]

Question 5.

- (a) The function $f(\theta, \phi)$ is an eigenfunction of the operator L_z with eigenvalue $m\hbar$, such that $L_+f(\theta, \phi) = 0$. Find the eigenvalue of the operator L^2 acting on the function $f(\theta, \phi)$. [6]
- (b) The radial part of the wave function in the first excited state of the Hydrogen atom ($2p$, $l = 1$, $m = 0$) is given by:

$$R_{21}(r) = Cre^{-r/2a}$$

where a is the Bohr radius. Find the normalization constant C and the most probable distance of the electron from the origin. [8]